Retrieval-Augmented Generation (RAG): A Practical Primer

Retrieval-augmented generation (RAG) combines a retriever with a large language model. At query time the system embeds the question, searches a vector database such as ChromaDB for semantically similar passages, and supplies those passages to the model as context. This grounds answers in source documents, reduces hallucination, and lets knowledge be updated without retraining. Typical failure modes include poor chunking, retrieval of irrelevant passages, and models that ignore the supplied context.

Evaluation of RAG should separate retrieval quality (recall@k, MRR) from generation quality (faithfulness, answer relevance). A frequent best practice is to build a small labelled question set and track both layers independently.

Multi-Agent Systems and Critic Loops

Multi-agent LLM systems divide work between specialised roles such as planner, researcher, critic and writer. A critic agent reviews intermediate output for unsupported claims and missing coverage, and can trigger additional retrieval before the final report is produced. Bounded iteration is important: without a maximum number of rounds, a critic can loop indefinitely and inflate cost.

Embedding models from the BGE family use an instruction prefix on queries but not on documents, and produce normalised vectors so cosine similarity equals the dot product.